Adaptive Knowledge Distillation for Lightweight Remote Sensing Object Detectors Optimizing

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***Abstract*— Lightweight object detector is currently gaining more and more popularity in remote sensing. In general, it is hard for lightweight detectors to achieve competitive performance compared with the traditional deep models, while knowledge distillation (KD) is a promising training method to tackle the issue. Since the background is more complicated and the object size varies extremely in remote sensing images, it will deliver lots of noise and affect the training performance when directly applying the existing KD methods. To tackle the above problems, we propose an adaptive reinforcement supervision distillation (ARSD) framework to promote the detection capability of the lightweight model. First, we put forward a multiscale core features imitation (MCFI) module for transferring the knowledge of features, which can adaptively select the multiscale core features of objects for distillation and focus more on the features of small objects by an area-weighted strategy. In addition, a strict supervision regression distillation (SSRD) module is designed to select the optimal regression results for distillation, which facilitates the student to effectively imitate the more precise regression output of the teacher network. Massive experiments on a large-scale Dataset for Object deTection in Aerial images (DOTA), object DetectIon in Optical Remote sensing images (DIOR), and Northwestern Polytechnical University Very-High-Resolution 10-class (NWPU VHR) datasets prove that ARSD outperforms the existing distillation state-of-the-art (SOTA) methods. Moreover, the performance of the lightweight model trained with our method transcends other classic heavy and lightweight detectors, which beneficiates the development of lightweight models.**

***Index Terms*— Adaptive knowledge distillation (KD), lightweight model, object detection, remote sensing imagery.**

I. Introduction

OBJECT detection plays a significant role in the realm of remote sensing and many R-convolutional neural network (CNN)-based approaches have been presented with the development of deep learning. However, the increase in remote sensing images from high-altitude satellites, drones, and airplanes has brought a lot of pressure on data transmission and processing, while the lightweight object detectors are capable of solving the above issue. The existing detectors are usually structurally complex, computationally expensive, and storage-consuming. These bring huge challenges to their deployment on intelligent terminals which lack computing resources. Recently, more light is shed on the design of lightweight detectors.

The existing lightweight detectors are usually unable to achieve competitive performance compared with classic heavy detectors. They are faced with some issues such as weak features' representation ability, high false alarm, and imprecise regression of bounding boxes. Some knowledge distillation (KD) methods, such as KD [1], FitNet [2], and AT [3], are proved to be valid to elevate the performance of the lightweight model. With these methods, the lightweight student model imitates the dark knowledge from the teacher model to promote its accuracy without extra costs. The core issue of distillation is which dark knowledge should be delivered. The existing solutions usually select the feature maps and regression results as the dark knowledge for distillation.

However, the above schemes still have the following issues:

Feature Imitation: The existing KD methods: Learning efficient object detection models with KD (LEOD) [4], Mimick [5], distilling object detectors with fine-grained feature imitation (FGFI) [6], and distilling object detectors with task adaptive regularization (TAR) [7] for object detection have been proposed recently. However, the above methods can only transfer the single highest level feature information, but they fail to propagate enough multiscale feature information to the student model since the sizes of objects vary greatly in remote sensing imagery. Furthermore, previous methods assign equal weight to the knowledge of small and large objects, while small objects in remote sensing imagery account for a very large proportion and only have few features in feature maps. As a result, small objects cannot be detected effectively (see Fig. 1).

Regression Distillation: The existing methods [4]–[7] only verify that distillation is capable of promoting the ability of the student model, but do not focus on optimizing the lightweight model. Since the overfitting during training, the lightweight model usually shows bad regression performance. Besides, these methods often leverage traditional KD for regression results and we will see a detection performance degradation if the student imitates all the regression results of the teacher model directly. For one thing, the above distillation methods

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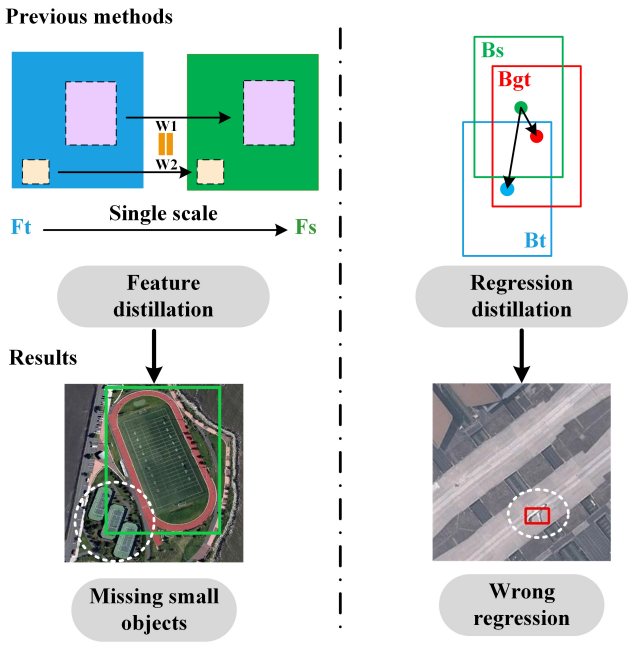


Fig. 1. Existing KD methods for object detection. In feature distillation, the blue and green regions represent the single-scale feature map of the teacher and student models, respectively, and the dotted boxes represent the feature deliver region with the same weight. In regression distillation, B_s, B_t, and B_{gt} denote the regression results of the student and teacher models, and ground truth, respectively. Then B_s imitates B_t and B_{gt} directly. Under single-scale feature equal weight distillation and misleading teacher regression distillation, poor results (white circle) may be obtained.

may give false guidance when the teacher's regression results are worse than the student's. For another, the student model tends to predict low-quality bounding boxes if the teacher's regression results are worse than the anchor's (see Fig. 1).

To deal with the issue of single-scale features' imitation, other researchers propose some multiscale imitation methods. However, they seldom select the core features of objects validly. They usually deliver multiscale redundant features of the whole image to the student model from the teacher model. The multiscale redundant features including massive backgrounds may weaken the features' representation ability of the student model. Therefore, a valid way to deal with this problem is to select the key features rather than all features as knowledge, which can alleviate the influence of background noise. Inspired by it, we propose a multiscale core features imitation (MCFI) module, which adaptively selects the multiscale core features of objects for imitation. Besides, an area-weighted strategy is designed to focus more on the features of small objects.

For the problem of all distillation regression results, others propose that the student model only imitates the positive sample regression results of the teacher model. However, not all the positive samples have a precise regression result and low-quality regression distillation will degrade the performance of the student model. Therefore, choosing appropriate regression results for distillation can avoid the issue effectively. Based on the above thoughts, the strict supervision regression distillation (SSRD) module is proposed to select optimal regression among them for distillation. Moreover, by adopting the location-weighted strategy, the student learns to predict high-quality regression outputs from the teacher model.

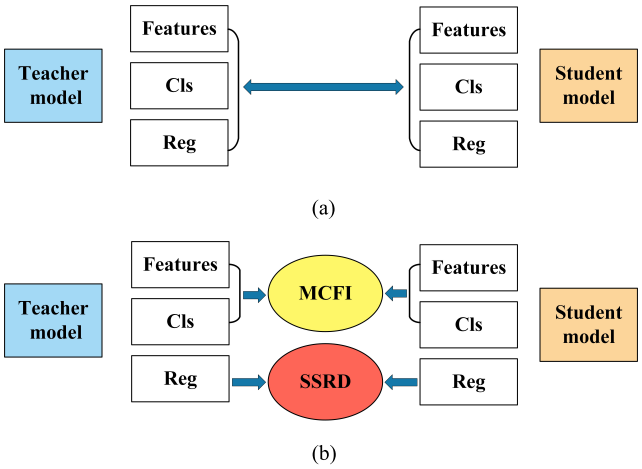


Fig. 2. Framework of KD for object detection. (a) In the previous method, the student directly imitates all knowledge of the teacher model. (b) In our method, the student imitates the core knowledge selected by the MCFI module and SSRD module from the teacher model.

The MCFI module and SSRD module are the key components of our proposed adaptive reinforcement supervision distillation (ARSD) framework, which validly tackle the lightweight object detectors' optimizing issue in remote sensing imagery. As illustrated in Fig. 2, the MCFI module and SSRD module transfer the selected knowledge of the teacher model. Massive experiments on three benchmark datasets [a large-scale Dataset for Object deTection in Aerial images (DOTA), Northwestern Polytechnical University (NWPU), and object DetectIon in Optical Remote sensing images (DIOR)] prove our method surpasses other state-of-the-art distillation approaches. Moreover, the performance of lightweight detectors trained with our method transcends other classic heavy and lightweight detectors.

The main contributions and innovations of this article are summarized as follows:

1) We propose a unified distillation framework called ARSD, which can tackle the lightweight object detectors' optimizing problem in remote sensing. The lightweight model trained with ARSD derives the best performance compared with other state-of-the-art (SOTA) distillation methods.
2) For the issue of feature imitation of remote sensing images, we propose an MCFI module, which adaptively selects the core information of objects for distillation. Besides, an area-weighted strategy is designed to focus more on small objects' information.
3) Aiming to solve the issue of regression distillation, we design an SSRD module, which assists the student to imitate the accurate regression outputs from the teacher model. Moreover, a location-weighted strategy is proposed to pay more attention to high-quality bounding boxes.

The remainder of the article is arranged as below. We introduce the relevant work in Section II. Our method will be elaborated in Section III. We present and analyze the results of experiments in Section IV. We sum up our job in Section V ultimately.

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II. Related Works

We will briefly review literatures which are relevant to our work including object detection and KD.

A. Object Detection

Modern CNN-based detectors, whether they are two-stage [8]–[11] or one-stage [12]–[15], consume large resources and need huge computation to reach desired performance, which hinders them from deploying on intelligent terminals without sufficient computing power. Commonly, the classical detectors mentioned above usually have a heavy backbone, such as Visual Geometry Group (VGG16) [16] and ResNet [17]. Therefore, some researchers focus on the design of lightweight backbone. MobileNet [18] builds lightweight deep neural networks with depthwise separable convolutions. Many lightweight detectors, such as MobileNet-single shot multibox detector (SSD) [18] and MobileNetV2-SSDLite [19], are designed when MobileNet is combined with one-stage detectors.

The previous detectors usually migrate the backbone designed for image classification task [16], [17] to detection task. However, the different characteristics between classification and detection tasks result in that lightweight backbone is not fit to directly migrate. Therefore, recent lightweight detectors Tiny-deeply supervised object detection (DSOD) [20] and Pelee [21] have designed specialized backbones. ThunderNet [22] proposes a compression backbone, region proposal network (RPN), and detection head. With the above modules, the detector achieves real-time detection on advanced RISC machine (ARM) platforms.

Compared with natural images, remote sensing imagery has the characteristics of complex background and multiscale objects, and the above lightweight detectors usually do not achieve satisfactory detection results. Therefore, some researchers have proposed many object detection algorithms based on deep learning for remote sensing images. Wang *et al.* [23] propose to adopt the atrous spatial feature pyramid (ASFP) component to fuse multiscale context information and use the loss weighted by area to make the model focus more on small objects. Sun *et al.* [24] propose to use the context-based deformable module (CBD) based on contextual information to combine attention and deformable convolution for rigid objects' detection. Wei *et al.* [25] propose to treat the task of aerial detection as the selection and clustering of the intersecting lines paired within the airplane. They also design a new pentagonal mask to aircraft representation, which can provide more accurate positioning. Xu *et al.* [26] put forward a novel pseudo anchor proposal module that solves the issue of spatial misalignment between ground truth and anchors. Cheng *et al.* [27] put forward a model which can learn rotation invariants, and they train the network by imposing new constraint rules, which can effectively deal with the problem of rotating objects. Cheng *et al.* [28] propose a discriminative objective function to boost the performance of remote sensing scene classification. Li *et al.* [29] design a local-contextual feature fusion network to promote the geospatial object detection. Cheng *et al.* [30] put forward a novel feature enhancement network for better object detection in remote sensing images.

B. Knowledge Distillation

Hinton *et al.* [1] first put forward KD for classification tasks. Usually, a teacher model is used to deliver dark information, which is the soft label of the output layer from the teacher model, to the student model. FitNet [2] proposes "hint learning" to distill the student model, where the hint is the information of the teacher's intermediate layers. AT [3] proposes that student mimics the attention maps of the teacher. Double similarity distillation (DSD) [31] proposes a double similarity distillation to promote the performance of the segmentation task. Relational KD (RKD) [32] proposes a relational KD method that transfers the distance and angle knowledge of sample pairs.

Although there is a lot of work about KD in classification tasks, it is difficult to directly migrate to more complicated detection tasks. LEOD [4] first proposes to use KD in object detection and the student model imitates each component of the teacher model. Mimick [5] proposes the student model to imitate the RoI region in the feature map of the teacher model, which contains negative and positive samples. FGFI [6] points out that the student model only imitates the regions close to the object in the teacher model's feature map. TAR [7] proposes that the student model imitates the teacher model's feature map of the ground-truth region, and only imitates the output result of the positive sample in the classification and regression heads. DeFeat [33] elucidates that features from different regions should be assigned different importance during distillation. Distilling knowledge via knowledge review (DKKR) [34] proposes the cross-stage connection paths for distillation. General instance distillation for object detection (GID) [35] puts forward a general instance selection module to make full use of all kinds of response distillation.

In recent years, some KD works have been introduced in remote sensing. Topological space KD (TSKD) [36] suggests a lightweight topological space network based on KD to extract roads, which can significantly improve the accuracy. Besides, they calculate the topological space loss with multiple depths and widths, which can heighten the weight of topological features. Bi-self-attention KD (SAD) [37] proposes an original self-attention supervision method for accurate and compact cloud detection which includes a forward inner distillation and a backward boundaries' distillation.

III. Methodology

Five parts will be presented in this chapter. They include problem setup, framework, MCFI module, SSRD module, and optimization.

A. Problem Setup

The task of KD is expressed as a knowledge delivery from the teacher model to the student model. It is usually realized by adding distillation loss. K represents the species of knowledge. T^k and S^k denote the knowledge of the teacher and student,

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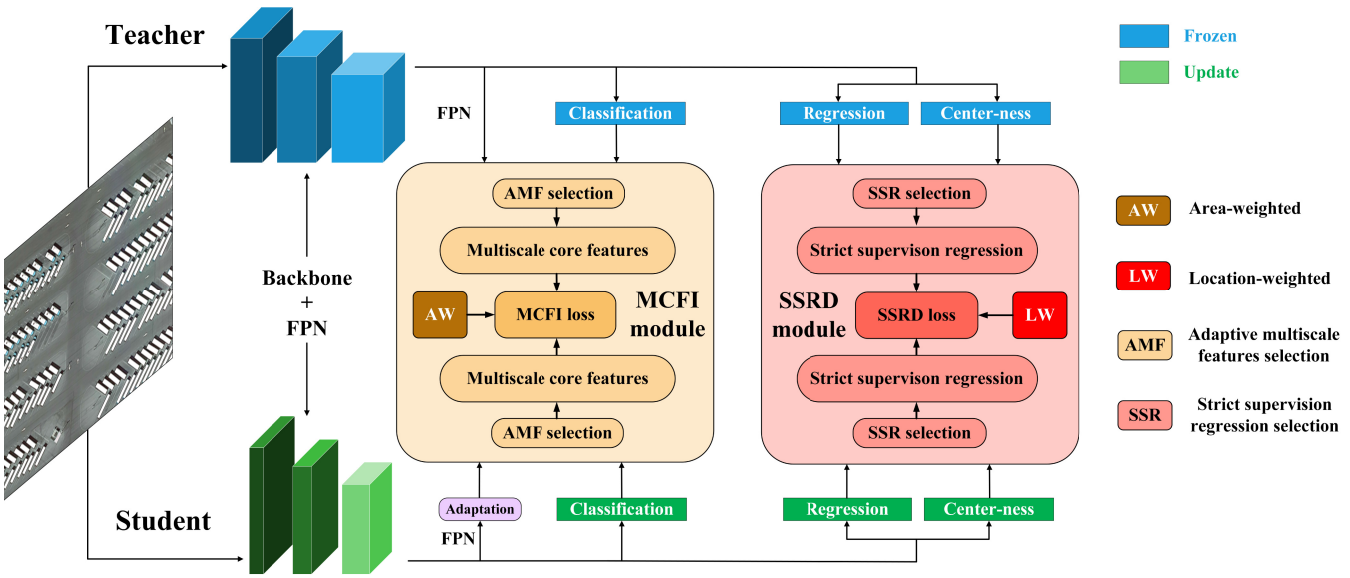


Fig. 3. Framework of our proposed ARSD, which comprises the MCFI module and the SSRD module. The MCFI module can be added in FPN or classification and the SSRD module can only be adopted to regression. It can adaptively select the dark knowledge and transfer it from the teacher model to the student model. During the training phase, only the parameters of the student model are updated, and the teacher model is frozen. **Adaptation** means the adaptation layers, which is illustrated in Fig. 4.

respectively. L_k represents the loss between them. The KD can be defined as follows:

$$\mathcal{L}_{\text{distillation}} = \sum_{k=1}^{K} \frac{\lambda_k}{N_k} L_k\left(T^k, S^k, R^k\right) \tag{1}$$

where λ_k represents the distillation weight and N_k denotes the normalization. Besides, R^k represents the regulation through which knowledge should be delivered. R^k of the previous methods usually deliver some noise or leave out some key knowledge, which makes the student model unable to imitate the core information from the teacher.

B. Framework

As shown in Fig. 3, we propose an ARSD framework which includes an MCFI module and an SSRD module. The student model imitates the multiscale core features' knowledge (feature maps of feature pyramid networks (FPNs) and classification) from the teacher model with the MCFI module, which can eliminate the noise of the background and focus more on features of small objects. The SSRD module transfers high-quality regression knowledge (regression results) which is selected by a strict supervision regression method. The SSRD module facilitates the student to learn to predict more precise regression outputs. Through the MCFI module and SSRD module, the elaborately selected core knowledge from the teacher is adaptively delivered to the student model. During the training phase, only the parameters of the student model are updated, and the teacher model is frozen.

The other modules of the framework are also shown in Fig. 3. Both the student and teacher models adopt the classic one-stage detector, which includes backbone, FPN, and detection head. We use ResNet as the backbone to extract multiscale features and fuse them by FPN. Then, features are sent to the detection head, which is divided into two branches. One branch is the classification head and the other is the regression head. A center-ness branch is parallel to the regression head simultaneously. Besides, to achieve a lightweight model, we reduce the number of student FPN layer channels to a quarter when conducting experiments on the small NWPU VHR-10 dataset. We reduce channels to half when we do experiment on the datasets of DOTA and DIOR. Besides, an adaptation layer is added to match the number of feature map channels as shown in Fig. 4.

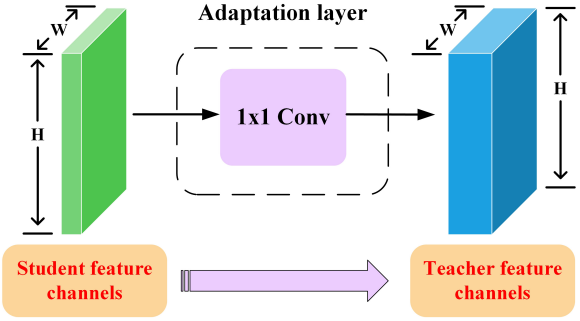


Fig. 4. Adaptation layer. The student feature map channels are equal to the teacher after the adaptation layer.

C. MCFI Module

In this section, the MCFI module will be presented in detail, which facilitates the student to imitate multiscale core knowledge of objects of the teacher model.

1) Feature Map Imitation: In the task of image classification, the student model usually imitates the entire feature maps of the teacher model. The L_2 loss is adopted between two feature maps to optimize the student model. But for the task of object detection, most regions are background in the feature maps. The student will learn noise and ignore the genuine object when imitating the feature map of the teacher model directly. Therefore, features should be elaborately selected so that the student only learns the object regions in the teacher feature maps. Taking these into consideration, we propose that the student should only imitate the feature maps of the ground-truth region because it is more likely to contain

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core knowledge. Motivated by the above thought, we propose an MCFI module that adaptively selects the multiscale core features of objects and pays more attention to the features of small objects.

2) Adaptive Multiscale Features Selection: Previous approaches transfer the single highest level feature or multiscale redundant features' information to the student model. Single-scale feature knowledge cannot validly deliver the information of objects of different sizes. The imitation of multiscale redundant features information including massive background may weaken the feature representation ability of the student model. Therefore, we first propose an adaptive multiscale features' selection method to choose the core information of diverse size objects from multiscale feature maps.

Given the input image $I \in R^{W\times H}$, where W and H represent the width and height of the image, respectively. Let $b^m = (x_1^m, y_1^m, x_2^m, y_2^m)$ indicate the coordinate of an object bounding box. $m \in [1, M]$, and M denotes the number of object in the image. We first calculate the area I_ϕ of the input image and compute the area b_ϕ^m of the mth object

$$\begin{aligned} I_\phi &= W * H \\ b_\phi^m &= \left(x_2^m - x_1^m\right) * \left(y_2^m - y_1^m\right). \end{aligned} \tag{2}$$

To estimate the information of each object belonging to scale feature maps, we calculate the target scale k_m according to its area as follows:

$$k^m = \max\left(\left\lfloor k_0 + \log_2\left(\sqrt{b_\phi^m / I_\phi}\right)\right\rfloor, 1\right). \tag{3}$$

We set k_0 to K, $k^m \in [1, K]$, and K represents the number of feature map scales. According to the above formula, the smaller the area of object, the lower scale features will be selected. The lower scale features usually have more information of small objects, which are more conducive to their detection.

Let $\mathcal{F} = \{F_1, F_2, \ldots, F_K\}$ represent a set of multiscale feature maps. Given a certain scale feature map $F_k \in R^{W_k\times H_k\times C}$, where W_k, H_k, and C denote the width, height, and channel numbers, respectively. The scaling ratio of certain scale feature map is calculated as follows:

$$S_k^w = \frac{W_k}{W}, \quad S_k^h = \frac{H_k}{H}. \tag{4}$$

Let $b_t^m = (x_{1_t}^m, y_{1_t}^m, x_{2_t}^m, y_{2_t}^m)$ denote the transformed coordinate of object on the corresponding scale feature map after scaling conversion, which are computed as follows:

$$\begin{aligned} x_{1_t}^m &= S_k^w x_1^m, \quad y_{1_t}^m = S_h^w y_1^m \\ x_{2_t}^m &= S_k^w x_2^m, \quad y_{2_t}^m = S_h^w y_2^m. \end{aligned} \tag{5}$$

With the transformed coordinate and target scale of the mth object, we generate a mask $\mathcal{M}^m \in R^{W_k\times H_k}$ that can highlight the features of the mth object and filter out the background

$$\mathcal{M}^m = \mathbb{1}\left[(i, j) \in b_t^m\right] \tag{6}$$

where $i \in [1, W_k]$ and $j \in [1, H_k]$. In mask $\mathcal{M}^m$, the value of the region where the object is located is 1, while the value of the region where the background is located is 0. We can get its core features $f_m \in R^{W_k\times H_k\times C}$

$$\begin{aligned} f_m &= \Upsilon\left(B_t^m, k^m, F\right) \\ &= \mathbb{1}\left[(i, j) \in b_t^m\right] * F_k \\ &= \mathcal{M}^m * F_k \end{aligned} \tag{7}$$

where $\Upsilon(\cdot)$ is our adaptive multiscale features' selection function.

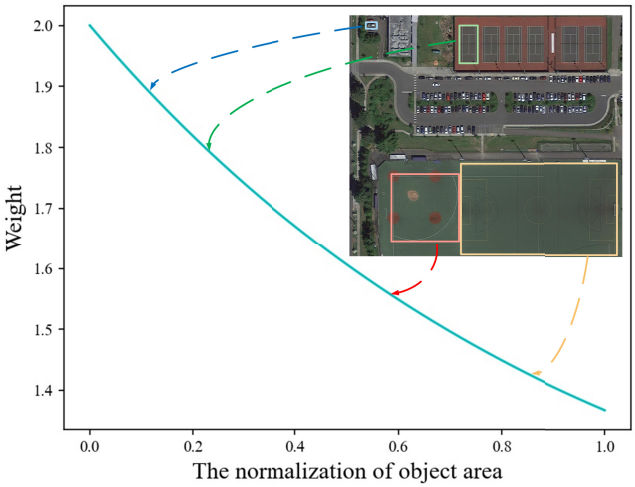


Fig. 5. Graph of area-weighted function. The distillation weight of the object decreases with the increase in area, which is used to enhance the distillation attention of small objects.

3) Area-Weighted Strategy: In the above part, we obtain the features of all the objects and the same imitation weight is given to all the sizes of objects. However, small objects in aerial images are harder to be detected because of the smaller proportion in feature maps. Therefore, we propose an area-weighted strategy to solve the above issue.

First, we calculate the normalized area of the mth object according to b_ϕ^m and I_ϕ as follows:

$$s_m = \sqrt{b_\phi^m / I_\phi}. \tag{8}$$

Second, we design a monotonically diminishing function that is related to the area of object. As illustrated in Fig. 5, an exponential function is adopted as the main function. As the size of an object decreases, it will have a greater weight

$$\omega_m = e^{-s_m} + 1. \tag{9}$$

Then, we get the area-weighted mask as below

$$\mathcal{M}^{m*} = \mathcal{M}^m * \omega_m. \tag{10}$$

To obtain the core features of all the objects in the image, we loop over all of them and get a set of mask $\mathcal{M} = \{\mathcal{M}^{1*}, \mathcal{M}^{2*}, \ldots, \mathcal{M}^{M*}\}$. We combine the same scale masks with OR operation. Finally, multiscale masks $\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_K\}$ are obtained.

By applying the multiscale masks $\mathcal{M}$, the student adaptively imitates the multiscale core information of the teacher model.

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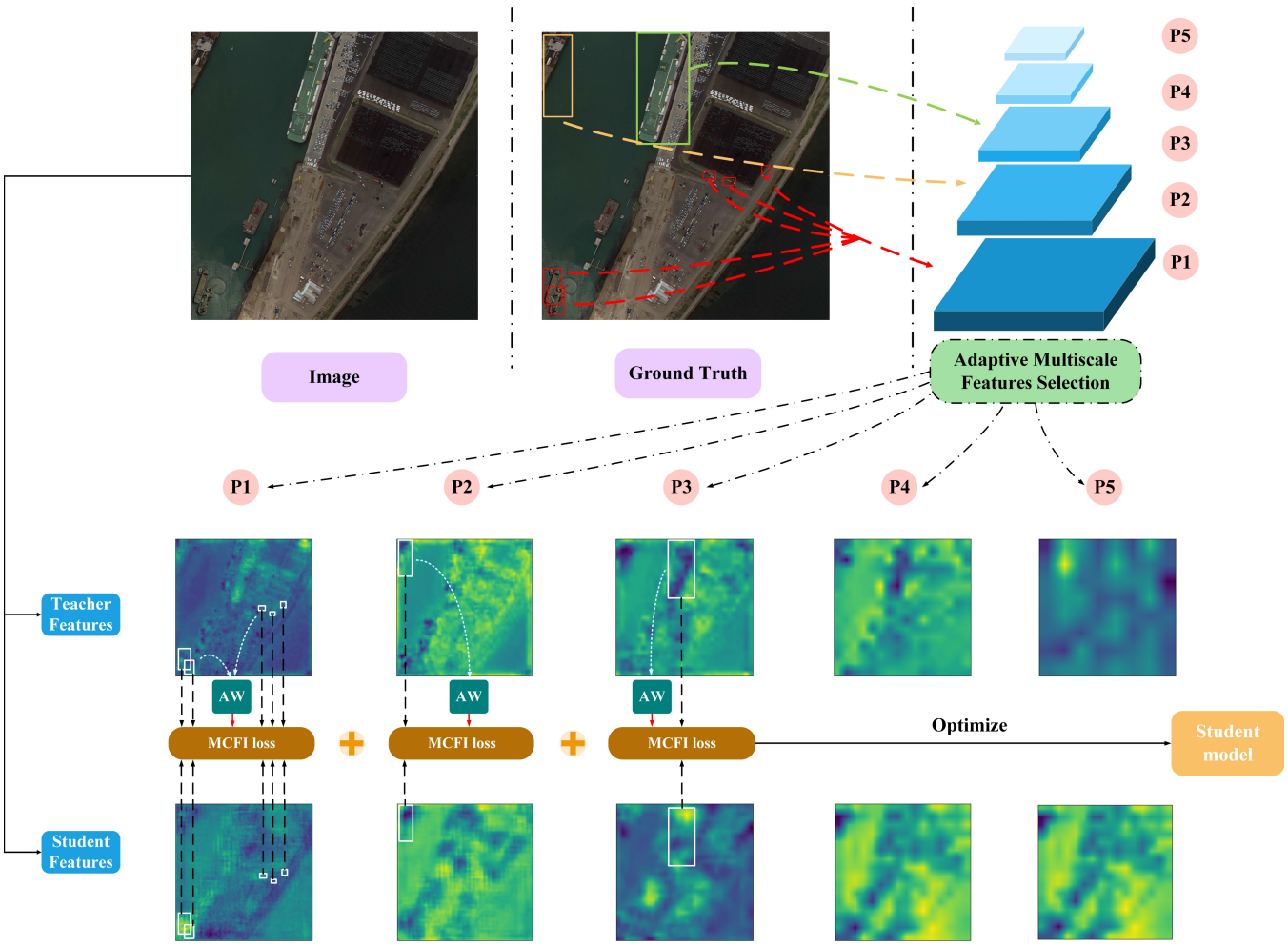


Fig. 6. Schematic scheme of the MCFI module. We adaptively select the multiscale core features of objects and the student only imitates the selected core features of the teacher model. The MCFI loss is calculated by all the core features between the teacher and student models. P1–P5 represent the five scale feature maps. Area-weighted (AW) is the area-weighted strategy, which is illustrated in Fig. 5.

The MCFI loss is defined as follows:

$$\mathcal{L}_{\text{MCFI}} = \sum_{k=1}^{K} \left[\frac{1}{2N_p^k} \sum_{i=1}^{W_k} \sum_{j=1}^{H_k} \sum_{c=1}^{C} \mathcal{M}_k \left(F_{ijkc}^t - F_{ijkc}^s \right)^2 \right] \tag{11}$$

$$N_p^k = \sum_{i=1}^{W_k} \sum_{j=1}^{H_k} \mathcal{M}_k \tag{12}$$

where W_k, H_k, and C are the width, height, and channels of feature maps, respectively. K represents the number of feature map scales. F_{ijkc}^s and F_{ijkc}^t denote the feature maps of the student and teacher models, respectively. N_p^k represents the amount of positive points in the mask $\mathcal{M}_k$.

The MCFI module is shown in Fig. 6.

D. SSRD Module

In this section, we describe the proposed SSRD module in detail, and the student model will learn more precise regression outputs from the teacher model with SSRD.

1) Strict Supervision Regression Selection: In the detection model, the regression head is responsible for predicting the coordinate offset of each anchor relative to its corresponding ground truth. Therefore, the regression head reveals the potential location of objects in an image. Since the features extracted by the teacher model are richer and the regression results are more accurate, we render the student model to imitate the regression head outputs from the teacher model. Although the regression head predicts the offsets of all the samples, only positive samples are used for supervision to avoid the influence of noise.

For a single-stage model, positive samples are the same when the student and the teacher models input the same image. B_t, B_s, B_a, and B_g are defined as decoded positive sample regression outputs of the teacher model, the decoded positive sample regression outputs of the student model, the positive anchors, and the ground truth, respectively. They are defined as follows:

$$\begin{aligned} B_\sigma &= \left\{ b_\sigma^1, b_\sigma^2, \ldots, b_\sigma^N \right\}, \quad \sigma = t, s, a \\ B_g &= \left\{ b^1, b^2, \ldots, b^M \right\} \end{aligned} \tag{13}$$

where N denotes the number of positive samples and M is the number of ground truth.

The teacher may give the student wrong guidance on some outputs when b_t is directly used to guide b_s. Therefore,

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we propose a strict supervision regression selection method to select the positive sample for high-quality distillation. It is necessary to define an evaluation criterion to measure the quality of the regression result and we define it as P. P of b_t^n, b_s^n, and b_a^n is calculated as below. IoU operation means calculating the overlap rate of the two sets of bounding boxes. max operation means getting the max IoU value between the prediction box and all ground truths

$$P_\sigma^n = \max\left(\mathrm{IoU}\left(b_\sigma^n, B_G\right)\right), \quad \sigma = t, s, a. \tag{14}$$

Besides, a map function $\Psi(\cdot)$ is proposed to get the index $\mathcal{R}$ of positive sample as follows:

$$\Psi(p) = \begin{cases} 1, & p > 0 \\ 0, & p \leqslant 0. \end{cases} \tag{15}$$

Adopting $\Psi(\cdot)$ with different selection methods, we will obtain different $\mathcal{R}$. The positive sample will be selected for distillation when $\mathcal{R} = 1$. We analyze different selection methods as below.

a) Weak supervision selection: The first selection method is similar to the strategy proposed in [7]. Positive sample is selected only when the prediction bounding box P of the teacher is greater than P of the anchor

$$\mathcal{R}_o^n = \Psi\left(P_t^n - P_a^n\right). \tag{16}$$

However, selecting such a sample for distillation makes the teacher mislead the student if the prediction bounding box of the student model is better than the teacher model.

b) Medium supervision selection: For the second selection method, we restrict the prediction bounding box P of the teacher to be greater than the student's

$$\mathcal{R}_m^n = \Psi\left(P_t^n - P_s^n\right). \tag{17}$$

But it may be possible that P of the student and teacher is not higher than P of the anchor. Such a selection method causes the student model to learn to predict the low-quality bounding box.

c) Strict supervision selection: The last is the strict supervision selection method

$$\mathcal{R}_s^n = \Psi\left(P_t^n - P_s^n\right) * \Psi\left(P_s^n - P_a^n\right). \tag{18}$$

Only when the above condition is satisfied can the student imitate the output of the teacher's prediction bounding box, which eliminates the disadvantages of the first two selection methods and render the student to learn more precise output from the teacher model.

2) Location-Weighted Strategy: In the previous part, we propose a strict supervision regression selection method to select suitable positive samples for distillation. However, the distillation weights of the selected samples are all equal, which is incapable of effectively distinguishing the location quality of different boxes. Driven by the above issue, we propose a location-weighted strategy to give the regression box with higher location quality greater distillation weight.

Let $b_a^n = \{x_1^n, y_1^n, x_2^n, y_2^n\}$ be a bounding box of positive sample before regression and $b^m = \{x_1^m, y_1^m, x_2^m, y_2^m\}$ denote its corresponding ground truth. Then we calculate the distance from the center of the anchor to the four sides of the ground truth

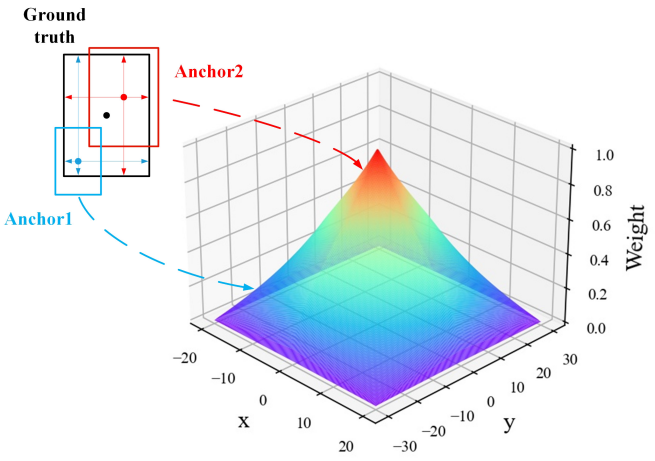


Fig. 7. Graph of location-weighted function. The origin of coordinates represents the center of the ground truth, and (x, y) denotes the center of the anchor. The weight of the anchor box decreases with the increase in the distance of the center point. The weight is used to enhance the distillation of the high-quality anchor.

$$l^n = \left|\frac{x_1^n + x_2^n}{2} - x_1^m\right|, \quad t^n = \left|\frac{y_1^n + y_2^n}{2} - y_1^m\right|$$
$$r^n = \left|x_2^m - \frac{x_1^n + x_2^n}{2}\right|, \quad b^n = \left|y_2^m - \frac{y_1^n + y_2^n}{2}\right|. \tag{19}$$

The location quality $\mathcal{Q}^n$ of b_a^n is calculated as follows:

$$\mathcal{Q}^n = \sqrt{\frac{\min(l^n, r^n)}{\max(l^n, r^n)} * \frac{\min(t^n, b^n)}{\max(t^n, b^n)}} \tag{20}$$

$$\mathcal{Q}^n \in [0, 1]. \tag{21}$$

With the center of the anchor nearer to the center of the ground truth, the value of $\mathcal{Q}^n$ gradually approaches 1 and the location quality of the box is higher, as shown in Fig. 7. Hence, we adopt $\mathcal{Q}^n$ as the weight for positive sample regression distillation, and the location-weighted selection method proposed by us is defined as follows:

$$\mathcal{R}_{\mathrm{sq}}^n = \mathcal{R}_s^n * \mathcal{Q}^n \tag{22}$$

looping over all the positive samples with the strict supervision selection method and we get the set $\mathcal{R} = \{\mathcal{R}_{\mathrm{sq}}^1, \mathcal{R}_{\mathrm{sq}}^2, \ldots, \mathcal{R}_{\mathrm{sq}}^N\}$.

By applying the index $\mathcal{R}$, only high-quality regression of positive samples is be distillated. The SSRD loss is defined as below

$$\mathcal{L}_{\mathrm{SSRD}} = \frac{1}{N_{\mathrm{reg}}} \sum_{n=1}^{N} \mathcal{R}^n * L_{\mathrm{giou}}\left(b_t^n, b_s^n\right) \tag{23}$$

$$L_{\mathrm{giou}}(b_t, b_s) = 1 - \frac{I(b_t, b_s)}{U(b_t, b_s)} + \frac{E(b_t, b_s) - U(b_t, b_s)}{E(b_t, b_s)} \tag{24}$$

$$N_{\mathrm{reg}} = \begin{cases} \sum_{n=1}^{N} \mathcal{R}^n, & \sum_{n=1}^{N} \mathcal{R}^n > 0 \\ 1, & \text{otherwise} \end{cases} \tag{25}$$

where L_{giou} represents the generalized intersection over union (GIOU) loss. $I(b_t, b_s)$ represents the intersection of the student

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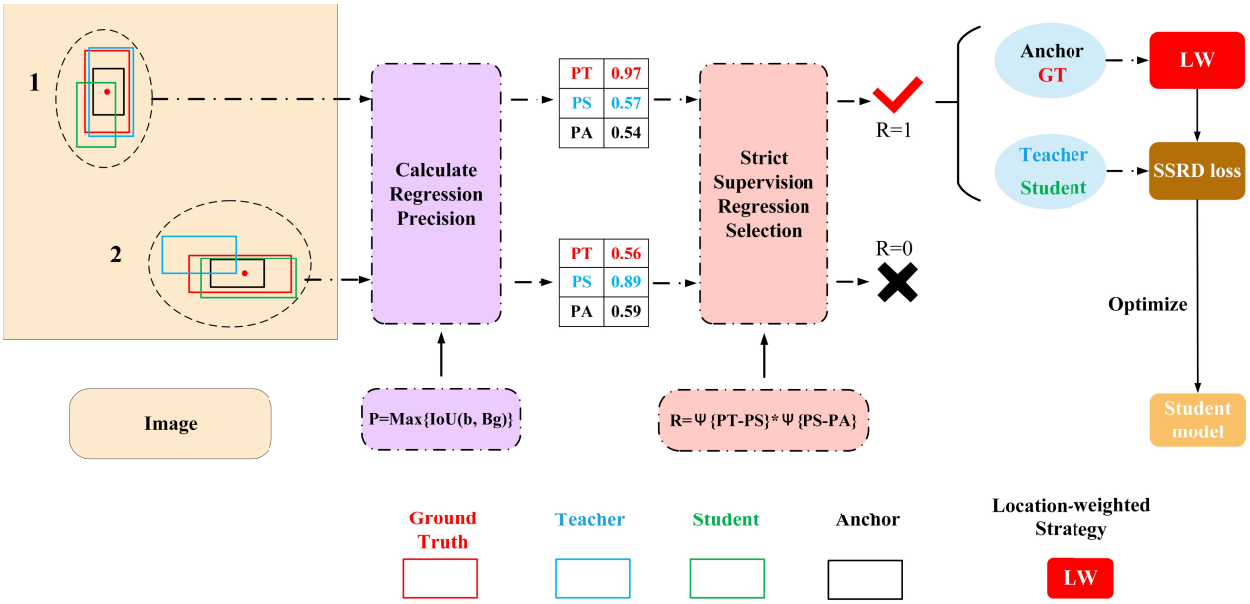


Fig. 8. Schematic of the SSRD module. Suppose there are two objects in the image, the red, blue, green, and black bounding boxes represent the ground truth, teacher model's prediction, student model's prediction, and anchor, respectively. We first calculate the regression precision P of the teacher, student, and anchor (P_t^n, P_s^n, P_a^n) using (14). Then, we adopt strict supervision regression selection to choose the optimal distillation samples (such as **1** in the image) according to (18). Finally, the SSRD loss is calculated by all the selected samples ($\mathcal{R} = 1$) between the regression results of the teacher and student models. Location-weighted strategy, which is illustrated in Fig. 7, is used to promote the high-quality regression box weight.

prediction box and the corresponding teacher prediction box. $U(b_t, b_s)$ represents the union of the student prediction box and the corresponding teacher prediction box. $E(b_t, b_s)$ represents the minimum enclosing box of the student prediction box and the corresponding teacher prediction box.

Fig. 8 elaborates our SSRD module in detail.

E. Optimization

Our ARSD framework includes the student and teacher models, which facilitates the student model to imitate core features' information and precise regression results of the teacher model. The parameters of the teacher model are frozen in the training phase, and only the parameters of the student model are updated. The student model is optimized by detection loss $\mathcal{L}_{\text{det}}$ and distillation loss $\mathcal{L}_{\text{distill}}$.

The detection loss is defined as follows:

$$\mathcal{L}_{\text{det}} = \lambda_1 \mathcal{L}_{\text{cls}} + \lambda_2 \mathcal{L}_{\text{reg}} + \lambda_3 \mathcal{L}_{\text{cen}} \tag{26}$$

where $\mathcal{L}_{\text{cls}}$ adopts focal loss, $\mathcal{L}_{\text{reg}}$ uses GIOU loss, and $\mathcal{L}_{\text{cen}}$ is a binary cross entropy (BCE) loss. We set $\lambda_1 = \lambda_3 = 1$ and $\lambda_2 = 2$.

The distillation loss is defined as below

$$\mathcal{L}_{\text{distill}} = \lambda_4 \mathcal{L}_{\text{MCFI}} + \lambda_5 \mathcal{L}_{\text{SSRD}}. \tag{27}$$

The whole loss is presented as below

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{distill}}. \tag{28}$$

The whole training process of the ARSD framework is illustrated in Algorithm 1.

Algorithm 1 ARSD

Input:

$D = \{X, Y\}$ is data for training, X, Y are images and their ground truths, respectively;

$\mathcal{T}$, θ_t are the teacher model and its parameters;

Output:

$\mathcal{S}$, θ_s are the student model and its parameters;

Load the weights of teacher model $\mathcal{T}$ and initialize the parameters of student model $\mathcal{S}$.

for every batch $D_b = \{X_b, Y_b\}$ **do**

  $\{F^t, B_t\} \leftarrow \mathcal{T}(X_b, \theta_t)$;

  $\{F^s, B_s\} \leftarrow \mathcal{S}(X_b, \theta_s)$;

  Get the multiscale masks $\mathcal{M}$;

  Get the regression indexes $\mathcal{R}$;

  $\mathcal{L}_{MCFI} = \mathcal{L}_{MCFI}(F^t, F^s, \mathcal{M})$;

  $\mathcal{L}_{SSRD} = \mathcal{L}_{SSRD}(B_t, B_s, \mathcal{R})$;

  $\mathcal{L}_{det} = \mathcal{L}_{det}(F^s, B_s, Y_b)$;

  $\mathcal{L} = \mathcal{L}_{MCFI} + \mathcal{L}_{SSRD} + \mathcal{L}_{det}$;

  Update the parameters θ_s of student model $\mathcal{S}$:

  $\theta_s = argmin_{\theta_s} \mathcal{L}$

end for

return $\mathcal{S}$, θ_s

IV. Experimental Results

In this part, the experimental details of our lightweight remote sensing object detectors' optimizing method will be presented. First, we present the datasets and implementation details. Then, we introduce the results and analysis of the experiment.

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TABLE I
TRAINING STRATEGY

Dataset	Image Size	Batch Size	Lr	Epoch
DOTA	1024×1024	8	0.005	30 (20,28)
NWPU	1000×600	4	0.001	40 (30)
DIOR	800×800	8	0.005	30 (20,28)

A. Datasets

1) DOTA: The DOTA dataset was proposed by Xia *et al.* [38] in 2018 for object detection of remote sensing. It contains 2806 large images and 188 282 instances, which are divided into 15 categories. The size of each original image is 4000 × 4000, and the images are cropped into 1024 × 1024 pixels with the overlap 200 pixels in the experiment. We select half of the original images as the training set, 1/6 as the validation set, and 1/3 as the testing set.

2) NWPU VHR-10: The dataset of NWPU VHR-10 was proposed by Chen *et al.* [27] in 2016. It contains 800 images, of which 650 pictures contain objects, so we use 520 images as the training set and 130 images as the testing set. The dataset contains 10 categories, and the size of the image is fixed to 1000 × 600.

3) DIOR: The DIOR dataset was proposed by Li *et al.* [39] in 2020 for the task of object detection, which involves 23 463 images and 192 472 instances. The size of each image is 800 × 800. We choose 11 725 images as the training set and 11 738 images as the testing set.

B. Implementation Details

1) Model Setting: Our teacher and student models adopt the classic one-stage model which is introduced in Section III-B.

Teacher Model: We use **ResNet101** as the backbone for the teacher on the datasets DOTA and DIOR. Since the NWPU dataset is small, **ResNet50** is used as the backbone for the teacher model.

Student Model: To design a lightweight model, the student uses **ResNet18** as the backbone. For the DOTA and DIOR datasets, the number of FPN channels is reduced to half. But for the NWPU dataset, we reduce its FPN channels to a quarter of the original.

2) Training Strategy: We also use different training strategies for different datasets, AND the detail is presented in Table I. We optimize the loss with stochastic gradient descent, the warm-up learning rate is set to 0.001, and the number of rounds is 500. The momentum and weight decay are set as 0.9 and 0.0001, respectively. The ImageNet pretrained model is used for backbone initialization to facilitate the network to converge better. In the test stage, the IoU threshold of nonmaximum suppression is 0.4 on DOTA and DIOR, and it is 0.6 on NWPU VHR-10. Our experiments are implemented on a Tesla V100 graphics processing unit (GPU).

3) Evaluation Metric: The mean average precision (mAP) is adopted to evaluate the accuracy of our model on the datasets. Before introducing AP, we briefly explain its relationship with TP, FP, FN, precision, and recall. The result is regarded as a true positive (TP) when the overlap ratio between the bounding box predicted by a detection result and the ground true box exceeds a certain threshold. Otherwise, it will be viewed as a false positive (FP). False negative (FN) will be labeled when a ground truth has no matching prediction results. The precision and recall are defined as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{29}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{30}$$

By sorting the predicted results according to the confidence, we can get different precision and recall, which can be drawn into a P-R curve, AP is the area below the P-R curve. We calculate the average value of all the categories' APs to get the mAP.

In addition, we adopt frames per second (FPS), floating point of operations (FLOPs), and amount of network parameters (Params) as indexes to evaluate model efficiency and lightweight performance.

C. Ablation Studies

In this section, we conduct the ablation experiments of our ARSD framework. We explore how each distillation module (MCFI and SSRD) promotes the performance of the student model. Besides, the experiments of hyperparameters' optimization, lightweight performance, and qualitative analysis are carried out. We conduct the ablation experiments on the dataset of DOTA.

1) ARSD Framework: The MCFI module can be applied to FPN layers or classification head. Whether MCFI is used alone or in combination, it can improve the accuracy of the lightweight model. The SSRD module can only be applied to the regression head and it also promotes performance. The ARSD means all the components are adopted simultaneously and the accuracy of the student model achieves 68.28% compared with the baseline of 63.8% as shown in Table II.

2) Effect of MCFI Module: We will obtain five scale feature maps after FPN layers and the student imitates different features from the teacher model according to various features' selection methods. As illustrated in Table III, FGFI is the fine-grained imitation in [6], $*$ means the threshold for feature selection is fixed for all scale features, while $\circ$ means the threshold is different. The two fine-grained imitations pay more attention to background information instead of foreground information. MCFI enables the student to learn about the features of the ground-truth region so that it can eliminate noise. $\star$ means the MCFI module without the area-weighted strategy. The last one is adding the area-weighted strategy, which facilitates the student model to pay close attention to small objects. As shown in Table III, the MCFI module accomplishes the optimal consequence and the detection accuracy has reached 66.95%, which increases by 3.15% mAP compared with the baseline.

We adopt FPN feature distillation and classification distillation at the same time, which use the same MCFI module. As shown in Table III, the simultaneous use of MCFI on two

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TABLE II

ABLATION STUDY OF ARSD FRAMEWORK. THE MCFI MODULE IS APPLIED TO FPN LAYERS OR CLASSIFICATION HEAD, AND THE SSRD MODULE IS ONLY APPLIED TO REGRESSION HEAD. ARSD REPRESENTS TWO MODULES ARE APPLIED AT THE SAME TIME

Method	FPN	CLS	REG	Pl	Bd	Br	Gft	Sv	Lv	Sh	Tc	Bc	St	Sbf	Ra	Ha	Sp	Hc	mAP
baseline	-	-	-	85.90	63.73	43.60	48.89	78.84	65.48	78.65	90.70	67.55	78.99	35.61	55.21	66.73	70.63	26.50	63.80
MCFI	√	-	-	87.12	66.71	48.15	55.40	80.21	67.05	79.05	90.67	72.10	78.31	33.89	60.09	72.07	72.54	40.86	**66.95**
	-	√	-	85.75	68.04	48.01	54.07	79.70	65.86	79.09	90.73	71.69	79.66	37.74	58.41	68.94	71.03	37.01	**66.38**
	√	√	-	86.71	67.90	47.27	57.35	80.44	67.10	79.16	90.79	73.38	79.51	38.19	60.23	71.81	72.12	41.81	**67.58**
SSRD	-	-	√	85.80	61.98	45.12	49.54	79.26	64.92	78.52	90.71	68.41	79.88	35.64	53.41	69.22	69.85	39.83	**64.81**
ARSD	√	√	√	86.90	69.64	46.38	56.85	80.60	66.96	78.96	90.76	72.15	78.96	39.67	61.27	72.23	72.39	50.64	**68.28**

TABLE III

ABLATION STUDY OF MCFI MODULE. $*$ MEANS THE FIXED THRESHOLD, $\circ$ REPRESENTS THE DYNAMIC THRESHOLD, $\star$ MEANS MCFI MODULE WITHOUT THE AREA-WEIGHTED STRATEGY

Method	FPN	CLS	baseline	FGFI*	FGFI$^\circ$	MCFI*	MCFI
MCFI	√	-	63.80	65.94	66.19	66.25	**66.95**
	-	√	63.80	65.83	65.97	66.17	**66.38**
	√	√	63.80	66.86	67.05	67.55	**67.58**

TABLE IV

ABLATION STUDY OF AREA-WEIGHTED STRATEGY FOR MCFI ON FPN

Method	function	mAP
MCFI	$2-s$	65.84
	$e^{-2s}+1$	66.41
	$e^{-s}+1$	**66.95**

regions achieves the best results compared with other methods while surpassing baseline of 3.78% mAP.

We conduct the area-weighted strategy distillation experiment on FPN features. We use a monotonically decreasing function as the weighting function. As the area of the object increases, the weight of its corresponding feature region decreases gradually. Table IV shows the exponential function surpasses the linear function among three kinds of weighting functions, and the basic exponential function reaches the best results.

According to the statistical results in [23], the bridge (Br), roundabout (Ra), helicopter (Hc), basketball court (Bc), baseball diamond (Bd), and soccer ball field (Sbf) have fewer instances. While most of Br, Ra, and Hc are small objects. As shown in Fig. 9, the growth ratio of Br, Ra, and Hc is bigger compared with Bc, Bd, and Sbf. Hence, it demonstrates the effectiveness of the MCFI module, which focuses more on small objects.

3) Effect of SSRD Module: After the MCFI module has been added to the classification and FPN, we also adopt regression distillation with three different selection methods mentioned above in Section III-D ($\mathcal{R}_o$, $\mathcal{R}_m$, $\mathcal{R}_s$). As shown in Table V, the strict supervision selection $\mathcal{R}_s$ has achieved the best result no matter what kind of loss functions are taken. Compared with $L1$ loss, GIOU loss reaches better results. After using

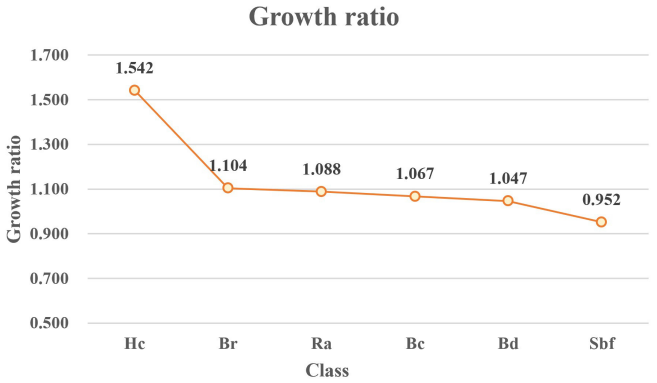


Fig. 9. Growth ratio of Br, Ra, Hc, Bc, Bd, and Sbf between our methods and baseline. They all have fewer instances, but Br, Ra, and Hc have more small objects. Results show that the MCFI module focuses more on small objects.

TABLE V

ABLATION STUDY OF SSRD MODULE ON REG. $L1$ AND GIOU REPRESENT THE OPTIMIZATION LOSS BETWEEN THE REGRESSION RESULTS OF THE STUDENT AND TEACHER MODELS. $\mathcal{Q}$ IS OUR LOCATION-WEIGHTED STRATEGY. $\mathcal{R}_o$, $\mathcal{R}_m$, AND $\mathcal{R}_s$ DENOTE DIFFERENT SUPERVISION SELECTION METHODS

Method	function	$\mathcal{R}_o$	$\mathcal{R}_m$	$\mathcal{R}_s$
SSRD	$L1$	67.44	67.56	**68.09**
	$GIOU$	67.79	67.73	**68.10**
	$GIOU+\mathcal{Q}$	67.79	67.90	**68.28**

GIOU loss and applying location quality $\mathcal{Q}$ at the same time, our SSRD module helps the model achieve the best performance, and the detection accuracy reaches 68.28% mAP, which is an increase of 4.48% compared with the baseline.

4) Hyperparameters Optimization: As presented in Section III-E, λ_4 anf λ_5 are the distillation weights of the MCFI and SSRD modules, respectively, so we test the accuracy of the distillation model under different weights which are shown in Table VI. We conduct the hyperparameter experiment on FPN feature distillation to find the best λ_4. As Table VI shows, the model reaches the best performance when $\lambda_4 = 0.01$. We also conduct the hyperparameter experiment on all distillation modules to find the best λ_5. As shown in Table VI, the model reaches the best result when $\lambda_5 = 0.1$.

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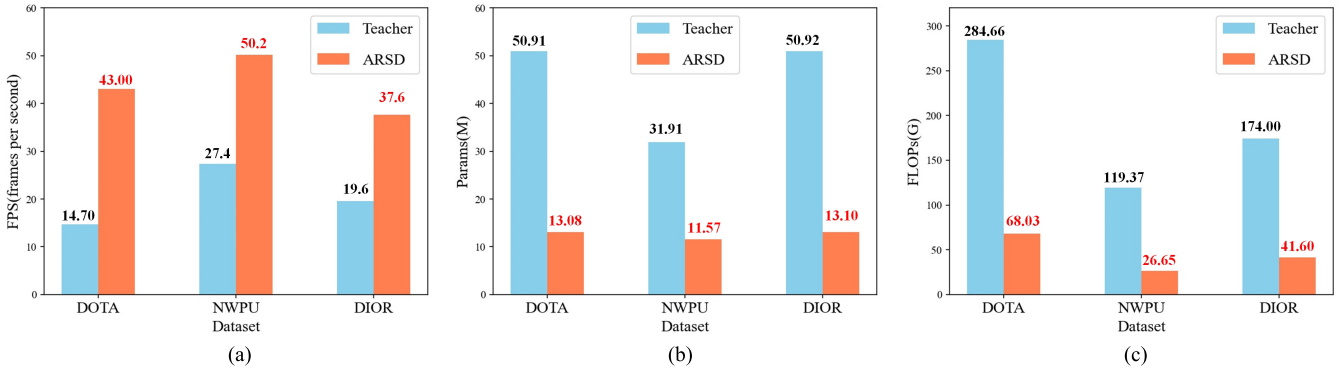


Fig. 10. Comparisons of the teacher model and lightweight model by FPS, parameters, and FlOPs on three datasets (DOTA, NWPU, and DIOR). The FLOPs and parameters of lightweight model are smaller, and the inference speed is faster. (a) FPS. (b) Params (M). (c) FLOPs (G).

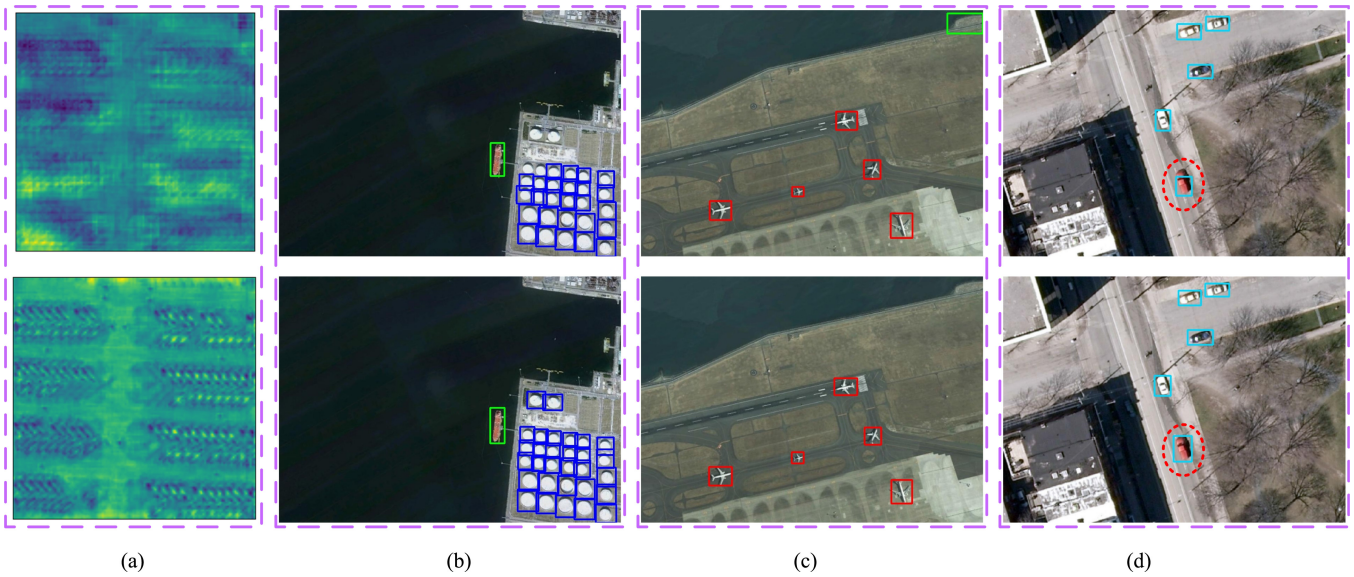


Fig. 11. Compared visualization results between the baseline and our ARSD model. The first row is the baseline and the second row is our ARSD model. Compared with the baseline, our model has a better feature extraction ability, a more accurate classification capability, and more precise regression effect. (a) Better features. (b) No miss. (c) No false alarm. (d) Precise regression.

TABLE VI
COMPARISONS OF DIFFERENT λ_4 AND λ_5 VALUES. "–" INDICATES THAT THE MODEL COULD NOT CONVERGE

λ_4	Weight	0.001	0.005	0.01	0.03	0.05
	mAP	65.65	66.4	**66.95**	-	-
λ_5	Weight	0.05	0.1	0.2	0.5	1
	mAP	68.14	**68.28**	67.41	68.19	54.41

5) Lightweight Analysis: Owing to the structure design, our student model is very lightweight. We compare the ARSD and the teacher model in terms of FPS, parameters, and FLOPs on three datasets. As Fig. 10 shows, ARSD has faster inference speed, smaller model parameters, and fewer FLOPs compared with the teacher model no matter in which dataset. Therefore, our lightweight model is more practical to be deployed on intelligent terminals.

6) Qualitative Analysis: As shown in Fig. 11, the first row is the visualization results of the baseline and the second row is the results of ARSD. Compared with the baseline, our ARSD model has a better feature extraction ability in Fig. 11(a), and the features of small objects are more evident. Besides, our model has no miss in Fig. 11(b) and no false alarm in Fig. 11(c). The detected car at the bottom in Fig. 11(d) demonstrates that our ARSD model has a more accurate regression result in comparison to the baseline.

D. Comparison With the State-of-the-Art Distillation Methods

To evaluate the efficacy of the ARSD distillation framework proposed by us, we show the competitive results between the six SOTA distillation methods and our method. For a fair comparison, we use the same student and teacher models as introduced in Section IV-B(1). As illustrated in Table VII, our ARSD transcends the existing distillation approaches on the DOTA, NWPU, and DIOR datasets, which proves the effectiveness of our ARSD framework.

1) DOTA: As shown in Table VII, KD [1] and FitNet [2] improve the accuracy of 2% and 2.5%, respectively. DeFeat [33] uses the background and foreground knowledge

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TABLE VII
COMPARISON WITH SOTA DISTILLATION APPROACHES ON THE DOTA, NWPU, AND DIOR DATASETS

	DOTA		NWPU		DIOR	
method	mAP	Δ AP50	mAP	Δ AP50	mAP	Δ AP50
teacher	71.65	-	93.21	-	71.7	-
student	63.80	-	86.56	-	65.2	-
KD [1]	65.8	+2	86.82	+0.26	67.2	+2
FitNet [2]	66.3	+2.5	86.85	+0.29	69.2	+4.0
FGFI [6]	66.25	+2.45	88.69	+2.13	69.5	+4.3
TAR [7]	68.09	+4.29	90.22	+3.66	69.5	+4.3
RKD [32]	67.19	+3.39	90.20	+3.64	67.3	+2.1
DeFeat [33]	64.11	+0.31	86.92	+0.36	67.7	+2.5
ARSD (ours)	**68.28**	**+4.48**	**90.92**	**+4.36**	**70.1**	**+4.9**

TABLE VIII
COMPARISON WITH CLASSIC DETECTORS ON THE DOTA, NWPU, AND DIOR DATASETS

	DOTA				NWPU				DIOR			
method	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)
FR-CNN [11]	60.64	7.0	60.19	289.25	77.80	25.4	41.17	127.70	54.1	16.8	60.21	182.02
RetinaNet [14]	50.39	14.0	55.39	293.36	89.40	26.3	36.29	123.27	65.7	18.2	55.49	180.62
YOLOv3 [40]	60.00	13.0	61.63	198.92	88.30	31.2	61.57	121.27	57.1	14.2	61.95	122.22
GFL [41]	66.53	23.8	19.13	159.18	88.80	34.7	19.13	91.73	68.0	36.5	19.13	97.43
FCOS [42]	67.72	18.3	31.57	202.15	89.65	24.4	31.86	116.63	67.6	25.4	31.88	123.51
ATSS [15]	66.84	26.6	18.97	156.01	90.50	37.4	18.96	89.90	67.7	26.7	18.98	95.50
MobileNetV2 [19]	56.91	22.5	10.30	124.24	76.9	30.2	10.29	71.49	58.2	30.1	10.32	76.1
ShuffleNet [43]	57.73	22.2	12.11	142.60	83.0	25.9	12.10	82.17	61.3	25.0	12.12	87.31
ARSD (ours)	**68.28**	**43.0**	**13.08**	**68.03**	**90.92**	**50.2**	**11.57**	**26.65**	**70.1**	**37.6**	**13.10**	**41.60**

for decoupled distillation, which results in the least increase in accuracy of 0.31%. RKD [32] adopts the relation distillation to improve 3.39% accuracy. FGFI [6] uses the fine-grained feature imitation which improves 2.45% performance. TAR [7] transfers the highest level features' knowledge with Gaussian mask, which promotes 4.29% performance. Compared with other approaches, our ARSD achieves the best improvement of **4.48%** in the dataset of DOTA.

2) NWPU: The NWPU dataset is a small dataset, and Table VII shows that our ARSD reaches the highest performance promotion of **4.36%**.

3) DIOR: We also conduct experiments on DIOR which has more categories compared with DOTA and NWPU. As presented in Table VII, KD [1], RKD [32], and DeFeat [33] improve the accuracy of the model to about 2%, while other methods bring the promotion of about 4%. Furthermore, our method realizes the optimal **4.9%** improvement.

E. Comparison With the Heavy Object Detectors

In this part, we compare our lightweight model with other classic heavy detectors including Faster R-CNN [11], RetinaNet [14], you only look once (YOLOv3) [40], generalized focal loss (GFL) [41], fully convolutional one-stage object detection (FCOS) [42], and adaptive training sample selection (ATSS) [15], as shown in Table VIII. Experiments on three datasets prove the efficacy of the ARSD framework. Not only does our lightweight model have higher accuracy but also it has a greater advantage in lightweight performance.

1) DOTA: Our ARSD achieves **68.28% mAP** compared with other detectors, surpassing the one-stage and two-stage detectors mentioned in Table VIII. In terms of inference speed, our ARSD has reached the highest speed of **43 FPS** compared with other detection models, while our model achieves the lowest model parameters (**13.08 M**) and FLOPs (**68.03 G**).

2) NWPU: We compare the results of our ARSD with other approaches on the dataset of NWPU. As illustrated in Table VIII, our ARSD achieves the best result (**90.92% mAP**) with the fastest inference speed (**50.2 FPS**), the smallest amount of model parameters (**11.57 M**), and the smallest FlOPs (**26.65 G**).

3) DIOR: As presented in Table VIII, our ARSD achieves the optimal detection result (**70.1% mAP**) and the fastest inference speed (**37.6 FPS**). In addition, the model parameters (**13.10 M**) and FLOPs (**41.60 G**) are much smaller than other SOTA detectors regardless of two-stage or one-

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Fig. 12. Three sets of visualization results from top to bottom are, respectively, on the (a) DOTA, (b) DIOR, and (c) NWPU datasets.

TABLE IX
GPU MEMORY OCCUPATION IN THE TRAINING PHASE

Methods	GPU memory (MB)		
	DOTA	NWPU	DIOR
MobileNetV2 [19]	14369	5155	9379
ShuffleNet [43]	21816	7555	14399
ARSD (ours)	5639	2485	3951

TABLE X
COMPARISON WITH DETECTORS DESIGNED FOR REMOTE SENSING REALM

Methods	DOTA			DIOR		
	mAP	FPS	Params	mAP	FPS	Params
O2-DNet [44]	71.1	15.28	209.0	68.3	7.63	209.0
FMSSD [23]	72.43	16	136.04	69.5	18.0	136.03
ARSD(ours)	68.28	43	13.08	70.1	37.6	13.10

stage. We also compare two detectors designed for remote sensing imagery such as feature-merged single-shot detection (FMSSD) [23] and oriented objects detection network (O2-DNet) [44]. As shown in Table X, their models have a close performance with our lightweight model. However, they usually have huger parameters and slower inference speed. Therefore, our model has a better balance.

Fig. 12 presents some visualization results of the DOTA, DIOR, and NWPU datasets from top to bottom. From the visualization results and above qualitative results, we can see that both large and small datasets have good detection results for objects of different scales.

F. Comparison With the Lightweight Object Detectors

In this part, we also compare our ARSD with other lightweight detectors such as MobileNetV2 [19] and ShuffleNet [43]. We replace the backbone of baseline with MobileNetV2 or ShuffleNet. As shown in Table IX, MobileNetV2 and ShuffleNet adopt the depthwise separable convolutions which occupy much GPU memory during the training phase and have larger FLOPs. As shown in Table VIII, although their model parameters are close to ours, their inference speed is slower than ours, and accuracies are

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about 10% less than ours. Therefore, our model has a better trade-off between performance and lightweight.

V. Conclusion

In this article, we propose an ARSD framework for lightweight object detectors' optimizing in remote sensing imagery. We design an MCFI module and an SSRD module, which transfer the selected dark knowledge from the teacher model to the lightweight student model. Our ARSD framework can promote the detection performance of the lightweight model without extra overhead. The experiments based on the DOTA, NWPU, and DIOR datasets certify that our ARSD achieves SOTA performance compared with other distillation methods. Moreover, the performance of our distillation lightweight model also surpasses other classic heavy detectors in both accuracy and inference speed, while model parameters and FLOPs are far smaller than others. Compared with other lightweight models, our method needs less training GPU memory, has faster inference speed, fewer FLOPs, and higher accuracy. Our ARSD has taken a significant step for the deployment of models in intelligent terminals with insufficient computing resources. In our future work, we will focus more on combining other compression methods with KD to further realize a more lightweight and high-performance model.

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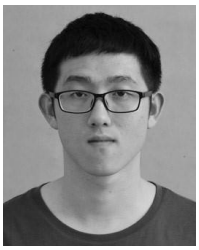
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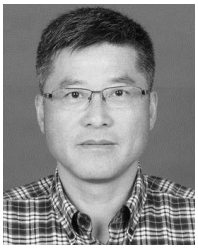
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